

Lecture Notes on NAND Flash Memory Technology

Chapter 7: 3D NAND Technology

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1 7.1 3D Architecture Principles

• Scaling Limitations of Planar NAND

Definition 1.1 (2D Scaling Challenges). Fundamental limits reached around 15-20nm:

- Cell-to-cell interference
- Short channel effects
- Quantum tunneling increase
- Manufacturing cost escalation

Scaling trend breakdown:

$$\text{Cost per bit} = \frac{C_{wafer} + C_{process}}{N_{die} \times N_{bit}} \quad (1)$$

As feature size F decreases:

$$N_{bit} \propto F^{-2}, \quad \text{but } C_{process} \propto F^{-n} \text{ where } n > 2 \quad (2)$$

• 3D Scaling Advantages

Density scaling with vertical integration:

$$\text{Density}_{3D} = \text{Density}_{2D} \times N_{layers} \times \eta_{stacking} \quad (3)$$

Cost reduction mechanism:

$$\text{Cost per bit}_{3D} = \frac{\text{Cost per bit}_{2D}}{N_{layers} \times \eta_{cost}} \quad (4)$$

where $\eta_{cost} \approx 0.7 - 0.8$ due to additional process steps.

2 7.2 Major 3D NAND Architectures

• BiCS (Bit Cost Scalable) - Toshiba

Theorem 2.1 (BiCS Structure). Key components:

- Vertical channel (poly-Si)
- Charge trap layer (SiN)
- Word line stacks (W)
- Bit line connections (top)

Memory hole diameter:

$$D_{hole} = D_{min} + 2 \times \text{CD variation} + 2 \times \text{Etch bias} \quad (5)$$

Channel resistance:

$$R_{channel} = \frac{\rho_{poly-Si} \times H}{A_{channel}} = \frac{\rho \times N_{layers} \times t_{layer}}{\pi(D_{hole}/2)^2} \quad (6)$$

• V-NAND (Samsung)

Vertical structure optimization:

$$I_{string} = \frac{\mu_{eff} C_{ox} W}{L_{eff}} \left(\frac{V_{GS} - V_{th}}{N_{layers}} \right) V_{DS} \quad (7)$$

Charge trap efficiency:

$$\eta_{trap} = \frac{N_{trapped}}{N_{injected}} = f(\text{SiN quality}, E_{field}, T) \quad (8)$$

• TCAT (Through Circuit Array Technology) - SK Hynix

Terabit Cell Array Transistor features:

- Vertical gate structure
- Through-silicon connections
- Advanced select gate design

Peripheral circuit integration:

$$A_{total} = A_{array} + A_{peripheral} \times \frac{1}{N_{layers}} \quad (9)$$

3 7.3 Manufacturing Challenges

- **High Aspect Ratio Etching**

Definition 3.1 (Aspect Ratio). Critical dimension control:

$$AR = \frac{H_{etch}}{D_{hole}} > 40 : 1 \text{ (typical)} \quad (10)$$

Etch rate uniformity:

$$\text{Uniformity} = \frac{\sigma_{etch}}{\text{mean}_{etch}} < 5\% \quad (11)$$

Etch selectivity requirements:

$$\text{Selectivity} = \frac{\text{Rate}_{target}}{\text{Rate}_{mask}} > 100 : 1 \quad (12)$$

- **Memory Hole Formation**

Multi-step etch process:

1. Hard mask definition
2. Through-oxide etch
3. Memory hole etch
4. Sidewall cleaning

Etch profile control:

$$\text{Taper angle} = \arctan\left(\frac{\Delta D}{2H}\right) < 1 \quad (13)$$

- **Layer Deposition Challenges**

Conformal deposition requirements:

$$\text{Conformality} = \frac{t_{sidewall}}{t_{field}} > 0.8 \quad (14)$$

Step coverage:

$$\text{Step coverage} = \frac{t_{min}}{t_{max}} \times 100\% \quad (15)$$

4 7.4 Process Integration

- **Gate Stack Formation**

Word line material considerations:

- **Polysilicon:** Low cost, high resistance
- **Tungsten:** Low resistance, difficult etch
- **TiN/W:** Barrier + conductor

Resistance-capacitance analysis:

$$\tau_{WL} = R_{WL} \times C_{WL} = \rho \frac{L^2}{A} \times \epsilon \frac{A}{t} \quad (16)$$

- **Channel Formation**

Poly-silicon channel properties:

$$\mu_{poly-Si} = \mu_0 \exp\left(-\frac{E_t}{kT}\right) \quad (17)$$

Grain boundary effects:

$$\mu_{eff} = \frac{\mu_{bulk}}{1 + \frac{qV_{gb}}{kT}} \quad (18)$$

- **Charge Storage Layer**

ONO vs. SONOS trade-offs:

$$\text{Retention} \propto \exp\left(\frac{E_{barrier}}{kT}\right) \quad (19)$$

Trap density optimization:

$$N_{trap,opt} = \frac{Q_{target}}{q \times V_{layer}} \quad (20)$$

5 7.5 Performance Characteristics

- **String Current Limitations**

Current degradation with layers:

$$I_{string}(N) = I_{single} \times \frac{1}{1 + \alpha N + \beta N^2} \quad (21)$$

Optimization strategies:

$$V_{pass,opt} = V_{th,avg} + \Delta V_{margin} + \frac{\alpha I_{target} R_{string}}{g_m} \quad (22)$$

- **Programming Characteristics**

Layer-dependent programming:

$$t_{program}(layer) = t_0 \times \left(1 + \gamma \times \frac{|layer - center|}{N_{layers}}\right) \quad (23)$$

Inter-layer interference:

$$\Delta V_{th,interference} = \sum_{i \neq target} \kappa_i \times \Delta V_{th,i} \quad (24)$$

- **Read Performance**

String resistance impact:

$$V_{cell} = V_{applied} - I_{string} \times R_{string} \quad (25)$$

Sensing optimization:

$$t_{sense} = \frac{C_{BL} \times V_{sense,min}}{I_{read,min} - I_{leakage}} \quad (26)$$

6 7.6 Design Optimization

• Layer Count Optimization

Optimal layer number:

$$N_{opt} = \arg \min \left(\frac{C_{process}(N)}{N \times Y(N)} \right) \quad (27)$$

where $Y(N)$ is yield as function of layer count.

• Memory Hole Diameter

Trade-off analysis:

$$D_{opt} = \min(D) \text{ subject to } I_{string} > I_{min}, Y > Y_{min} \quad (28)$$

Capacitive coupling:

$$C_{coupling} = \frac{2\pi\epsilon_0\epsilon_r}{\ln(D_{outer}/D_{inner})} \quad (29)$$

• String Architecture

Select gate optimization:

$$N_{SG} = \lceil \log_2(N_{strings}/BL) \rceil \quad (30)$$

Dummy cell placement:

$$N_{dummy} = f(\text{Edge effects, Programming uniformity}) \quad (31)$$

7 7.7 Advanced 3D Concepts

• Multi-Deck Architecture

Deck separation advantages:

$$\text{Yield}_{total} = \prod_{i=1}^{N_{decks}} \text{Yield}_i \quad (32)$$

Process complexity reduction:

$$\text{AR}_{deck} = \frac{\text{AR}_{monolithic}}{N_{decks}} \quad (33)$$

• Advanced Materials

High-k dielectric integration:

$$C_{storage} = \epsilon_0\epsilon_r \frac{A}{t} \quad (34)$$

Metal gate benefits:

$$R_{gate} = \rho_{metal} \frac{L}{W \times t} < 10\Omega/\square \quad (35)$$

• Scaling Roadmap

Future scaling paths:

$$\text{Density}_{future} = \text{Density}_{current} \times 2^{generation} \quad (36)$$

Technology node progression:

- 64-layer $\rightarrow$ 96-layer $\rightarrow$ 128-layer $\rightarrow$ 176-layer
- Diameter scaling: 90nm $\rightarrow$ 80nm $\rightarrow$ 70nm
- String length: 3m $\rightarrow$ 8m $\rightarrow$ 12m

8 Examples

Example 8.1 (String Resistance Calculation). Calculate total string resistance for: - 128 layers, hole diameter: 80nm - Poly-Si resistivity: 10 cm - Layer thickness: 60nm

Solution:

$$A_{channel} = \pi \left(\frac{80 \times 10^{-9}}{2} \right)^2 = 5.03 \times 10^{-15} \text{ m}^2 \quad (37)$$

$$L_{total} = 128 \times 60 \times 10^{-9} = 7.68 \times 10^{-6} \text{ m} \quad (38)$$

$$R_{string} = \frac{10 \times 10^{-2} \times 7.68 \times 10^{-6}}{5.03 \times 10^{-15}} \quad (39)$$

$$= 1.53 \times 10^7 = 15.3 \text{ M} \quad (40)$$

High resistance requires optimization.

Example 8.2 (Cost Analysis). Compare cost per bit for 2D vs 3D: - 2D: 20nm node, 4F² cell size - 3D: 40nm node, 64 layers, 6F² cell size

Solution:

$$\text{Area}_{2D} = (20 \times 10^{-9})^2 \times 4 = 1.6 \times 10^{-15} \text{ m}^2/\text{bit} \quad (41)$$

$$\text{Area}_{3D} = \frac{(40 \times 10^{-9})^2 \times 6}{64} = 1.5 \times 10^{-16} \text{ m}^2/\text{bit} \quad (42)$$

3D provides 10.7 $\times$ density improvement.

Assuming 30% process cost increase for 3D:

$$\text{Cost ratio} = \frac{1.3}{10.7} = 0.12 \quad (43)$$

3D reduces cost per bit by 8 $\times$ despite process complexity.

9 Homework Problems

Assignment 7.1: 3D NAND Technology

1. Architecture Comparison:

- Compare BiCS, V-NAND, and TCAT

- Analyze density, performance, cost
 - Design optimization strategies
 - Predict future evolution paths
- 2. Manufacturing Process Design:**
- Design 128-layer etch process
 - Calculate aspect ratio requirements
 - Optimize selectivity and uniformity
 - Include process margin analysis
- 3. String Performance Modeling:**
- Model current vs. layer count
 - Include temperature effects
 - Optimize pass voltage scheme
 - Design compensation algorithms
- 4. Layer Scaling Analysis:**
- Model yield vs. layer count
 - Calculate optimal deck architecture
 - Analyze cost-density trade-offs
 - Design for 512-layer target
- 5. Memory Hole Optimization:**
- Optimize diameter vs. performance
 - Model etch profile effects
 - Design CD control strategy
 - Include corner rounding effects
- 6. Advanced Materials Integration:**
- Design high-k storage dielectric
 - Model metal gate implementation
 - Analyze thermal stability
 - Optimize work function engineering
- 7. Interference Analysis:**
- Model layer-to-layer coupling
 - Design compensation schemes
 - Optimize programming algorithms
 - Include read disturb effects
- 8. Peripheral Integration:**

- Design under-array circuitry
- Optimize charge pump placement
- Model thermal effects
- Design for testability

9. Reliability in 3D:

- Model layer-dependent wear
- Design adaptive algorithms
- Analyze failure mechanisms
- Optimize for automotive applications

10. Future Technology Concepts:

- Design 1024-layer architecture
- Explore novel materials (2D, CNT)
- Model quantum effects at scale
- Propose beyond-charge storage

Next Chapter Preview: Chapter 8 will cover "Emerging Technologies and Applications" including beyond charge storage, storage class memory, and AI accelerator applications.